

Aerospace Altitude & Speed Profile Analyzer

Christopher Bourjaili, Dakota D'Angelo, and Benjamin Oakes

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Problem Overview / System Description

What system are you modeling?

- A flight simulation that tracks both speed and altitude at given time intervals between 0-5 hours

Why is this system important in engineering?

- Helps predict arrival times and ideal conditions for flight

Real-world application

- Physics: Kinematics showing variables such as velocity, time, and displacement

Engineering Objective

What are you trying to analyze or predict?

- In this, we are trying to integrate a system where it tracks different phases of flight targeting altitude and speed

What question does your model answer?

- What are the specific times that have transitions from different phases?
- What's the relationships between speed and altitude with time?

Mathematical Model

Present the governing equation(s)

Linear Climb/Descent: $\text{Altitude} = mt + b$ (similar to $y = mx + b$)

Cruise: $\text{Altitude} = A + B \sin(Ct)$ (Simulates wobbles at cruise height)

Rate of Change: $\text{Rate} = (\Delta \text{Altitude}) / (\Delta \text{Time})$ (Determines if plane is climb/descent)

Define all variables and constants

- Variables: t , Altitude, Speed
- Constants: 35,000ft , 450kts
- Time Step: Each time interval is 30 seconds apart

Algorithm / System Logic

Initial Conditions

Time: Flight starts at origin 0 seconds, and goes by increments of 30 seconds converted into hours

- **Represented as:** $t_hours = (0:30/3600:5)'$;

Altitude: Flight starts at zero ft (ground level)

- **Threshold constant created:** $CRUISE_ALT_THRESHOLD = 30000$;

Speed: Flight starts at zero

Algorithm / System Logic

Update Rule

- The setup of this code highly depends on time
 - Altitude updates as time changes every 30 seconds over 5 hours
 - Speed updates as time changes every 30 seconds over 5 hours

Algorithm / System Logic

If /else statements

- These if/else statements creates certain actions to happen under the conditions that are set

`if altitude(i) > CRUISE_ALT_THRESHOLD` : This creates an If statement to show the cruise phase at 30,000 ft or greater

`phases(i) = "Cruise";`

`elseif alt_rate(i) > 500` : This checks the altitude rate if it is greater than 500 ft/hr

`if speed(i) < 200` : This creates the takeoff phase and climb if speed is less than 200, if speed is greater than 200 its in climb phase

`phases(i) = "Takeoff";`

`else`

`phases(i) = "Climb";`

`end`

`elseif alt_rate(i) < -500` : This is the altitude rate at which displays the descent phase if plane descends faster than 500 ft/hr

`phases(i) = "Descent";`

`else`

`if altitude(i) < 1000` : If plane is less than 1000ft this is considered as takeoff, however, if plane is greater than 1000 ft but not changing altitude significantly, label cruise.

`phases(i) = "Takeoff";`

`else`

`phases(i) = "Cruise";`

MATLAB Implementation

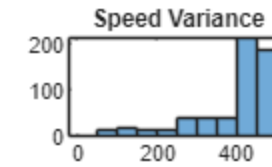
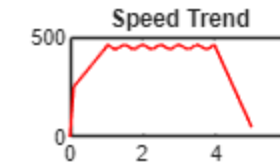
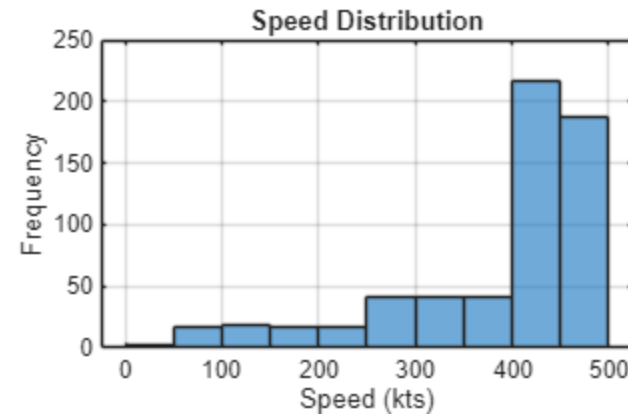
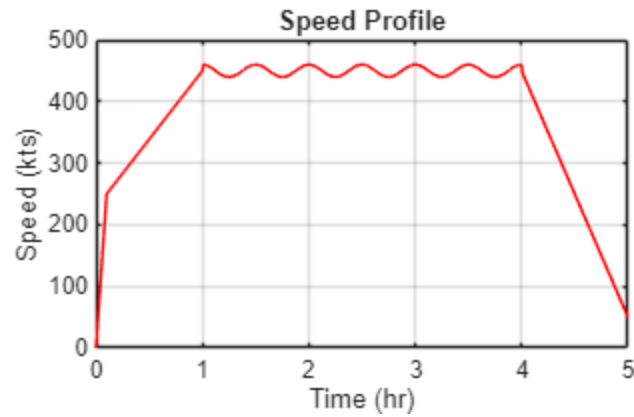
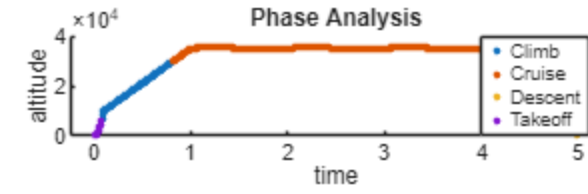
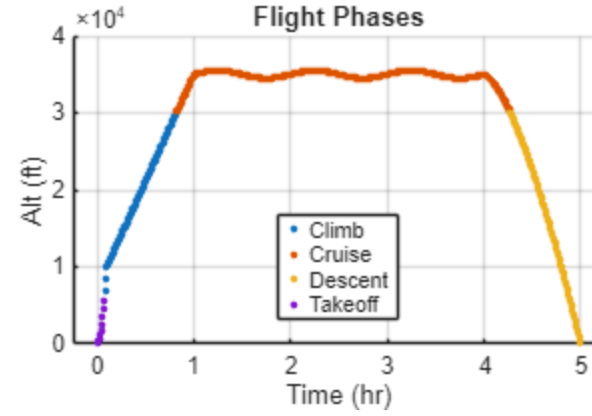
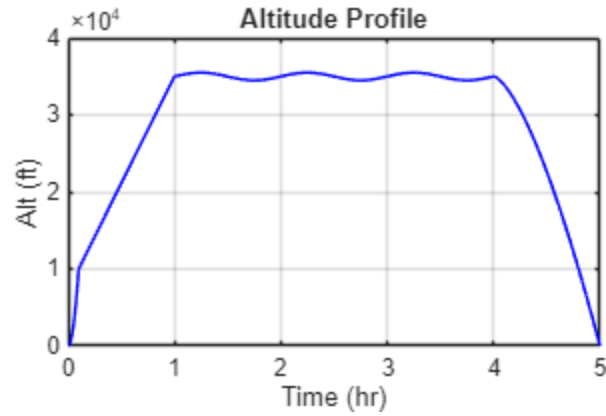
```
t_hours = (0:30/3600:5)';
```

```
CRUISE_ALT_THRESHOLD = 30000;
```

```
altitude_ft = zeros(size(t_hours));  
speed_kts = zeros(size(t_hours));
```

```
for i = 1:length(time)  
    if altitude(i) > CRUISE_ALT_THRESHOLD  
        phases(i) = "Cruise";  
    elseif alt_rate(i) > 500  
        if speed(i) < 200  
            phases(i) = "Takeoff";  
        else  
            phases(i) = "Climb";  
        end  
    elseif alt_rate(i) < -500  
        phases(i) = "Descent";  
    else  
        if altitude(i) < 1000  
            phases(i) = "Takeoff";  
        else  
            phases(i) = "Cruise";  
        end  
    end  
end
```


Simulation Results (Plots)



Analysis of Results



What is happening in the graph?



Does it make sense physically?



Identify:

Trends
Steady-state behavior
Threshold effects

Analysis of Results

What is happening in the graph?

In 5 of these graphs, they have different relationships:

- **Altitude vs Time** graph Displays Altitude Profiles and Phase Profiles
- **Speed vs Time** graph Displays Speed Profiles
- **Frequency vs Speed** Graph that displays the speed distribution

Analysis of Results

Does this make sense physically?

- Shows realistic scenarios of a typical flight
 - **First hour:** Goal is to reach the cruise height
 - **Last Hour:** Descending back down sea level (landing)
 - **Speeds:**
 - At takeoff phase realistically range from 200 kts
 - At cruising altitude speeds realistically range at 450 kts

Analysis of Results

Trends:

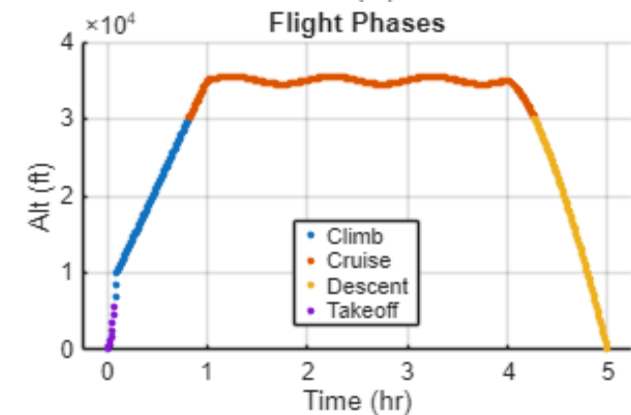
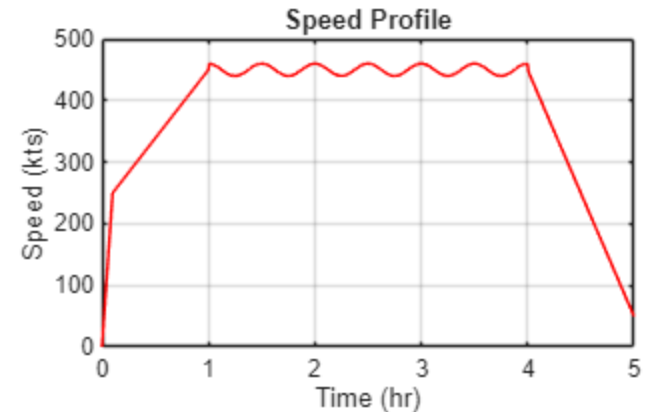
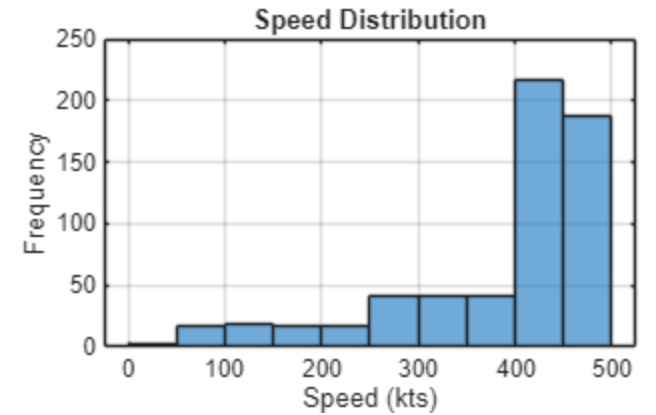
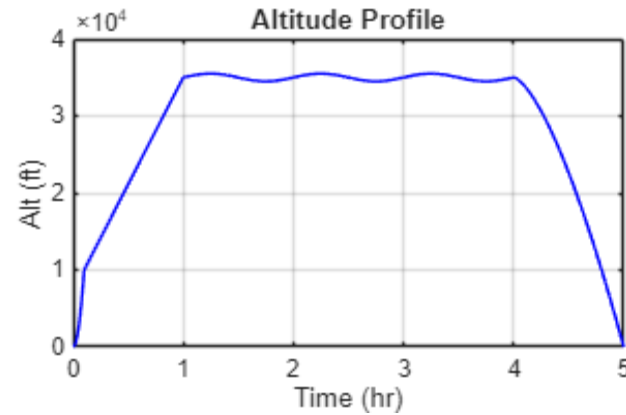
- Time is a major variable that controls our altitude and speed depending on the time of hour of the flight

Steady-State Behavior

- At each phase of the flight exempting the cruise, each traveled in a linear pattern

Threshold effects

- Because of the if/else statements created the speed and altitude change the phases of flight while the threshold for altitude is 30,000 ft



Parameter Sensitivity / Exploration

- **What happens if you change a parameter?**
 - **Cruise Threshold**
 - Changing threshold changes the target height
 - **Example:** Changing 30,000ft to 35,000 ft changes the output of the graphs

Model Limitations

What assumptions did you make?

- Flight climbing and descending "perfectly"
- Didn't account for wind resistance

What does your model NOT capture?

- Fuel Consumption
- Doesn't account for outside forces
 - **Example:**
 - Lift
 - Drag

Engineering Relevance

Why does this matter in real engineering?

- Allows for efficiency; an optimal flight path and plan
- Safety to ensure planes stay within a safe area depending on ascent/descent speeds and altitude limitations
- **Where could this model be used?**
 - This is a great representation of what is shown at airports
 - Allows to display flight path and estimation of when the plane will land

Team Contributions

Member 1 (Dakota D'Angelo): Worked majority on presentation and partial of coding

Member 2 (Benjamin Oakes) Worked majority on writing code and partially on presentation

Member 3 (Christopher Bourjaili) Worked on creating notations/comments on lines of code and partially on presentation

Conclusion

What was created?

- Created a flight simulator simulating different phases (Takeoff, Climb, Cruise, and Descent) while displaying altitude and speed given at time intervals

What did we learn?

- Mathematical equations that were used to create visuals for graphs
- Learned if/else statements

Key Takeaways

- Flight simulator created to have predictions for flight durations based on speed and time
- Multiple graphs created to show visuals of different phases and showing speed and altitude meters